CNCC

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2026中国计算机大会

China National Computer Congress 2026



成都 | 2026年 10月21—24日

动态图表征学习：语义

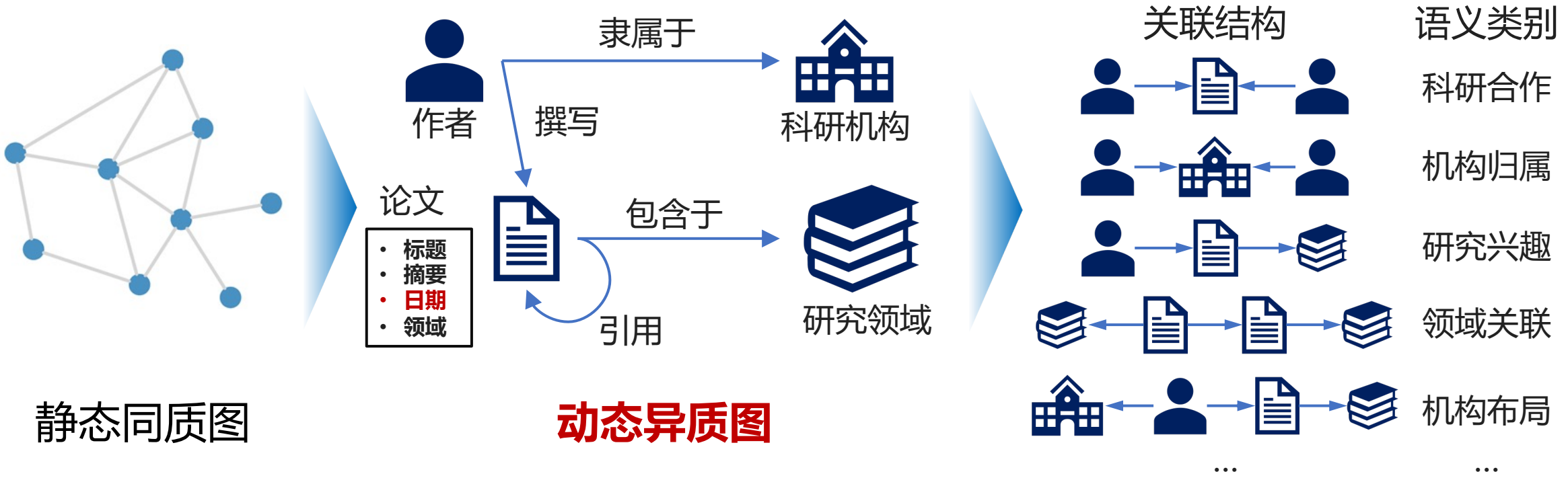
刘欢 杭州电子科技大学

内容大纲

- 研究背景与意义
- 研究问题与挑战
- 概念定义
- 方法分类
- 总结与展望

动态异质图

- 现实世界系统：**多类型**主体及交互，随时间**动态演化**
- 语义演化： 由实体及关系组合表达的动态关联含义
- 动态异质图：通用表征形式，建模图语义演化的关键范式

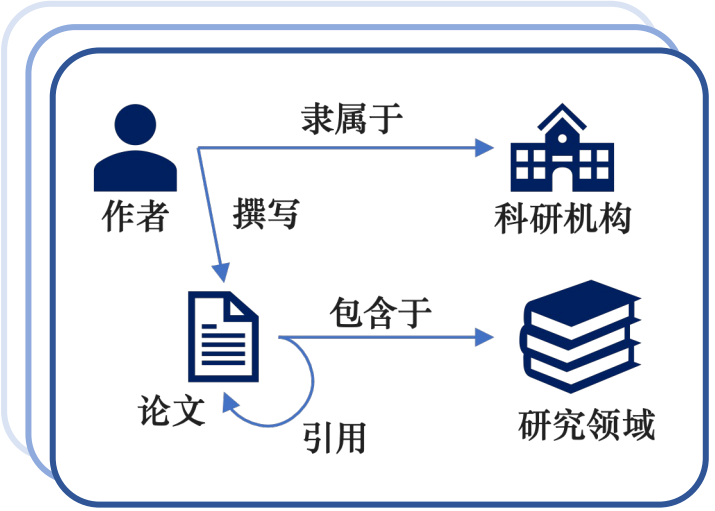


数学定义及符号表示

➢动态异质图 $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{X}, \mathcal{A}, \mathcal{R}, \mathcal{T}, \phi, \varphi)$,

- 节点集 $\mathcal{V}$
- 边集 $\mathcal{E}$
- 属性矩阵 $\mathbf{X}$
- 节点类型集 $\mathcal{A}$
- 边类型集 $\mathcal{R}$
- 时间域 $\mathcal{T}$
- 节点类型映射 $\phi : \mathcal{V} \rightarrow \mathcal{A}$
- 边类型映射 $\varphi : \mathcal{E} \rightarrow \mathcal{R}$

满足 $|\mathcal{T}| > 1 \wedge |\mathcal{A}| + |\mathcal{R}| > 2$

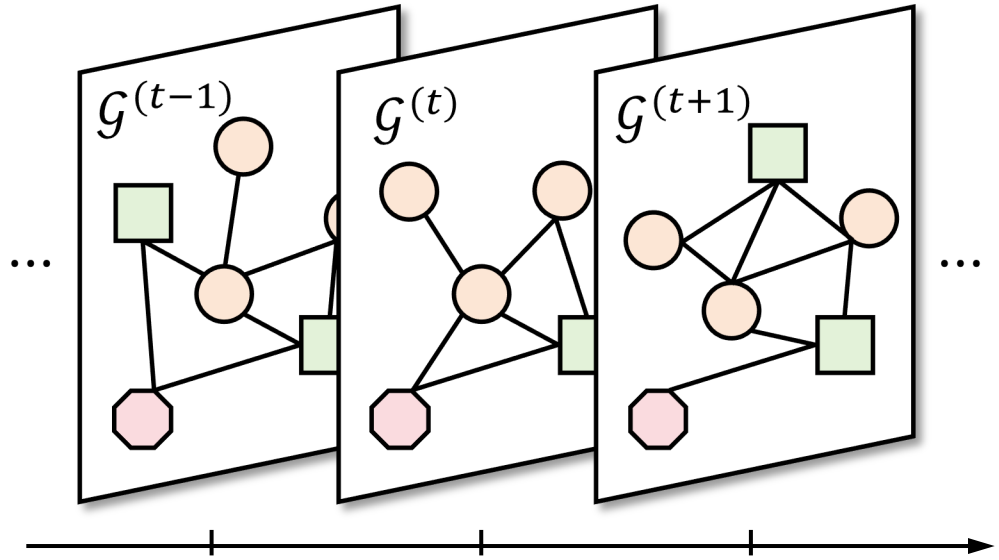


➢与其他类型图的关联:

时间 \ 类型	单一语义	多类型语义
静态关联	静态同质图	静态异质图
时序演化	动态同质图	**动态异质图**

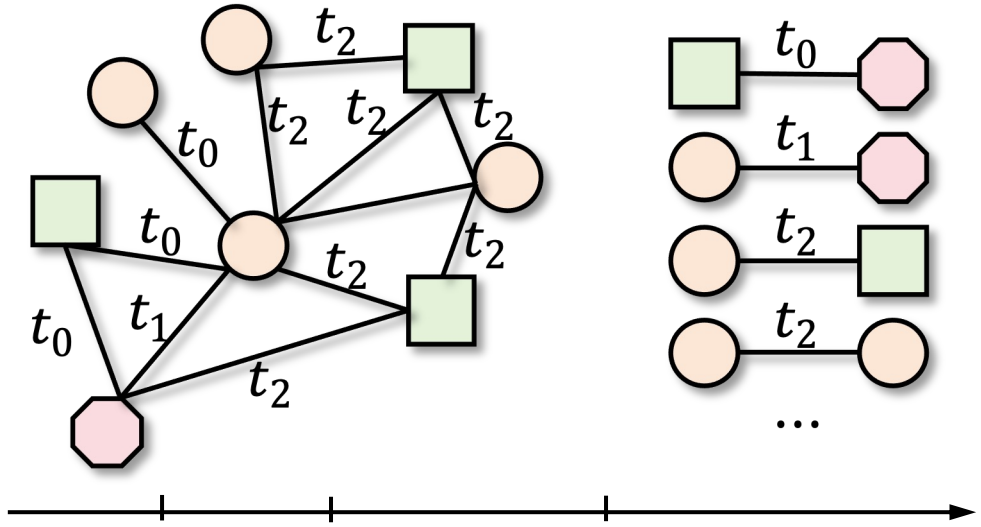
动态异质图快照与事件流

- 根据时间域 $\mathcal{T}$ 是否均匀，可划分为**离散时间**与**连续时间**动态异质图



$$G^{(t)} = (\mathcal{V}^{(t)}, \mathcal{E}^{(t)}, \mathbf{X}^{(t)}, \mathcal{A}, \mathcal{R}, \phi, \varphi)$$

离散时间动态异质图

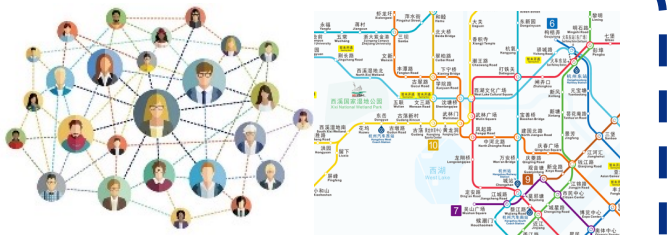


$$\mathcal{E} = \{(u, v, r, t) \mid u, v \in \mathcal{V}, r \in \mathcal{R}, t \in \mathcal{T}\}$$

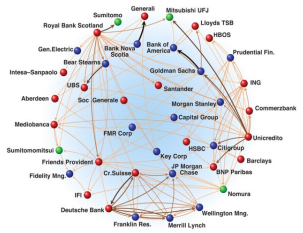
连续时间动态异质图

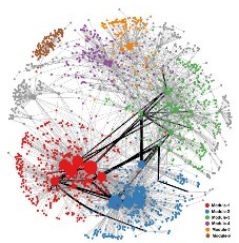
动态异质图表征学习

- 输入：动态异质图 $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{X}, \mathcal{A}, \mathcal{R}, \mathcal{T}, \phi, \varphi)$
- 输出：保留**异质语义**与**时间依赖**的低维表征 $f_\Theta : \mathcal{G} \to \mathbb{R}^{|\mathcal{V}| \times d}, d \ll |\mathcal{V}|$

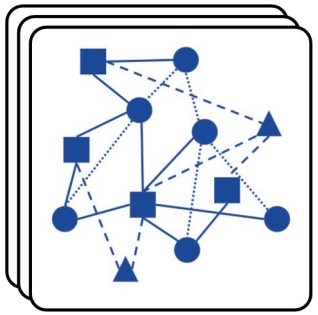

社交网络

交通网络


金融网络


基因调控网络

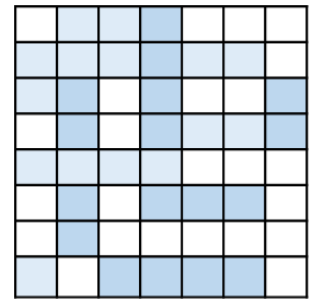
现实世界复杂系统


动态异质图

$\mathcal{G} = \{G^{(t)}\}_{t=1}^{T}$

$f : u \to \mathbb{R}^d$

可训练映射


嵌入向量 $\boldsymbol{Z}$

动态异质图表征学习

任务

节点分类

链路预测

异常检测

应用

虚假账号检测

异常交易识别

信息传播预测

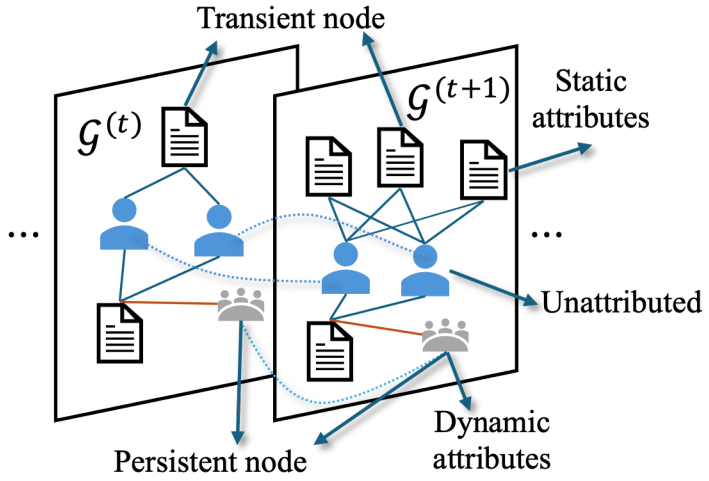
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下游任务

研究问题与挑战

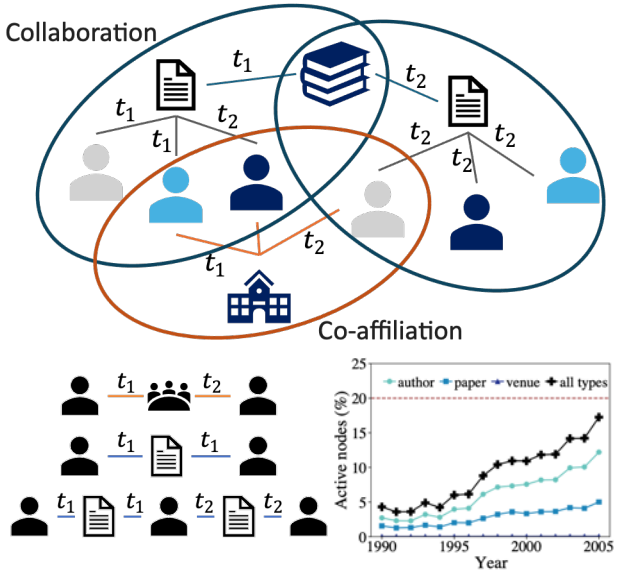
- 研究问题：如何建模现实网络中多类型语义关联的异步演化？实现可计算、可迁移、可理解的上下文表征学习？

微观



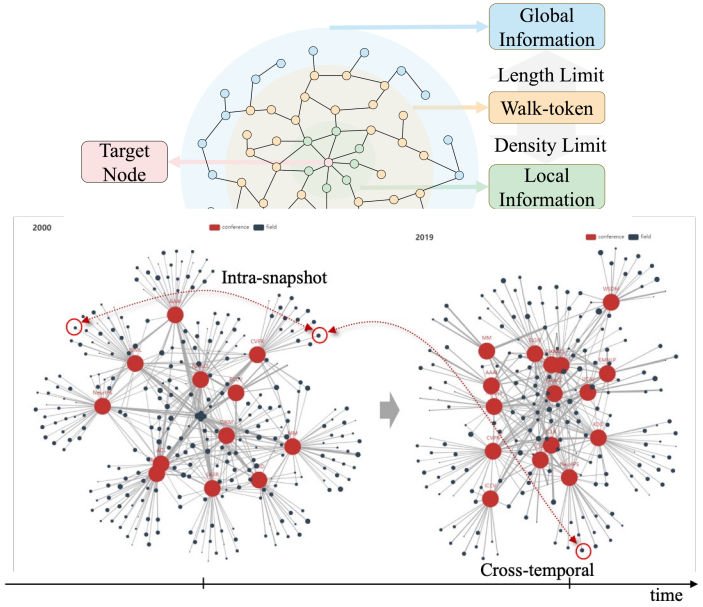
节点语义易漂移

介观



关系演化难建模

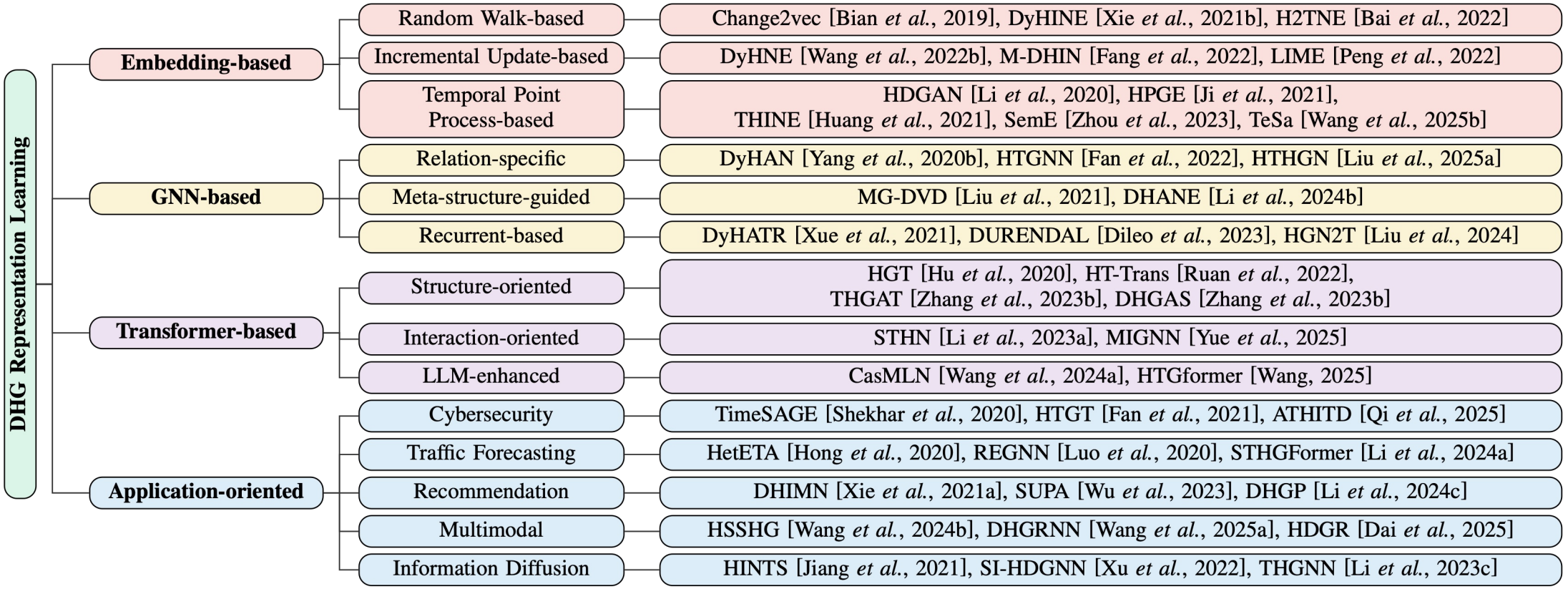
宏观



长程依赖难计算

动态异质图表征学习：方法体系

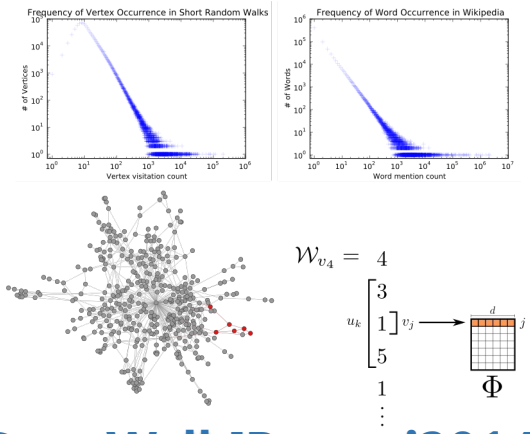
- 以算法为中心的面向动态异质图的表征学习方法分类体系



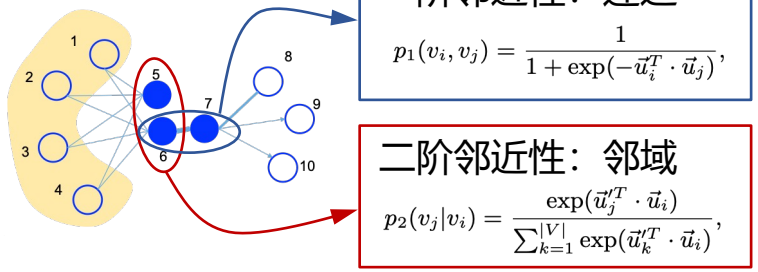
动态异质图表征学习分类体系[1]

基于图嵌入的表征学习

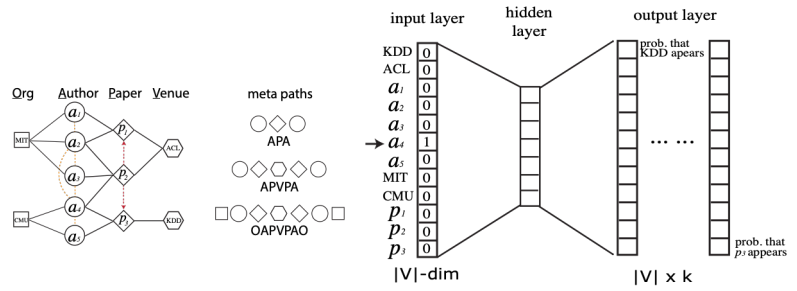
- 定义：直接优化低维节点向量，保留结构邻近性



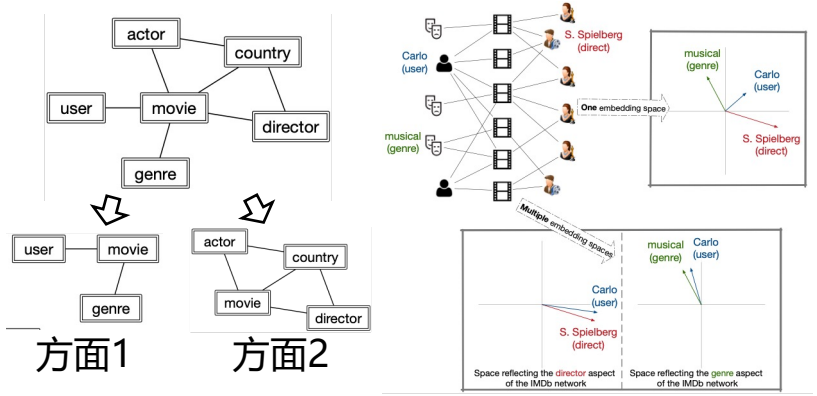
DeepWalk[Perozzi2014]



LINE[Tang2015]



metapath2vec[Dong2017]



AspEm[Shi2018]

- ❖基于随机游走
  - DHNE[Yin2019]
  - H²TNE[Bai2022]
- ❖基于增量更新
  - M-DHIN[Fang2022]
  - DyHNE[Wang2022]
  - LIME[Peng2022]
- ❖基于时序点过程
  - HPGE[Ji2021]
  - THINE[Huang2021]
  - SemE[Zhou2023]

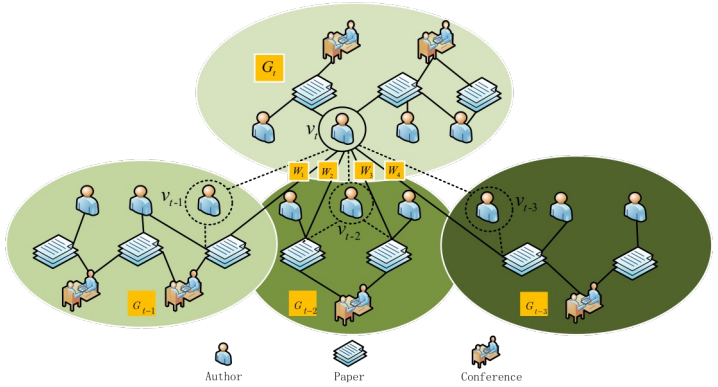
基于随机游走的图嵌入

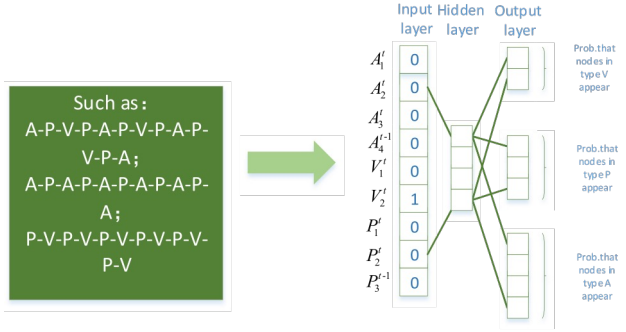
- 动机：动态语义上下文可通过随机游走序列化建模

$$\mathcal{L} = \sum_{v \in V^{(t)}} \sum_{c \in \mathcal{N}_S^{(t)}(v)} \log P(c \mid v; \theta),$$

$\mathcal{N}_S^{(t)}(v)$：随机游走采样得到的时间感知语义邻居

DHNE[Yin2019]





1、为历史交互设定时序衰减权重或更新策略：

$$W(N_{t_j}|N_{T_i}) = \frac{exp[-(T_i - t_j)]}{\sum_{T_i - \tau < t_m < T_i} exp[-(T_i - t_m)]}$$

历史交互时间差

2、候选游走节点满足语义类型约束，如元路径/元图：

$$P(v_{i+1}|v_i^c, M) = \begin{cases} \dfrac{W(v_{i+1}|v_i^c)}{\sum_{v_j \in H_{c+1}(v_i^c)} W(v_j|v_i^c)}, & (v_{i+1},\ v_i^c) \in E,\ f(v_{i+1}) = c+1 \\ 0, & (v_{i+1}, v_i^c) \notin E \text{ or } f(v_{i+1}) \neq c+1 \end{cases}$$

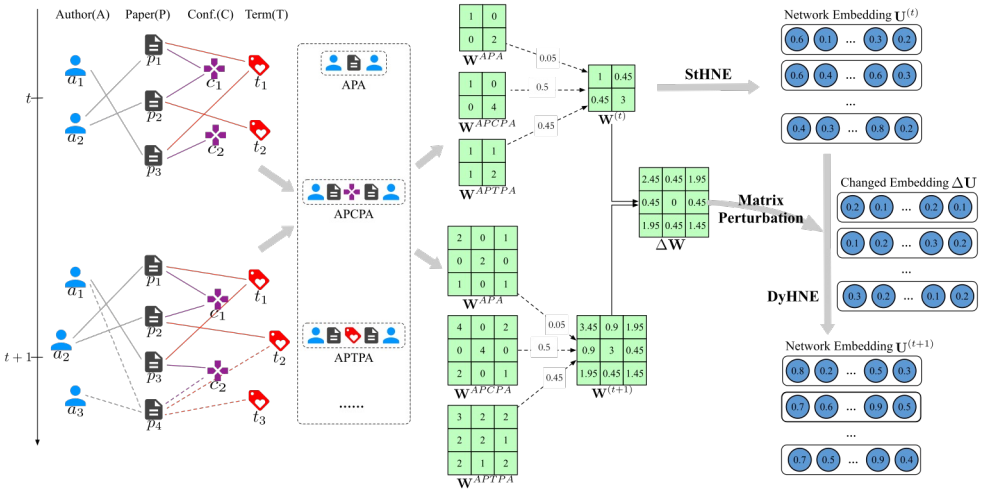
3、根据转移概率采样游走序列，通过skip-gram框架优化

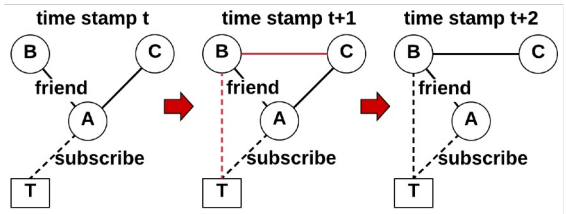
[1] Yin, Ying, et al. DHNE: Network Representation Learning Method for Dynamic Heterogeneous Networks. IEEE Access 2019.

基于增量更新的图嵌入

- 动机：网络语义平滑演化，变化可视为微小扰动

$$\mathbf{h}_u^{(t)} = f_{\text{upd}}\left(\mathbf{h}_u^{(t-1)}, \Delta\mathcal{G}^{(t)}\right)$$





Change2vec[Bian2019]
M-DNIH[Fang2022]
对三元组开放与闭合节点执行动态更新

DyHNE[Wang2022]

1、针对不同语义构造元路径增强矩阵，同时保留两阶邻近性：

$$\mathcal{L}_1^m = \sum_{v_i, v_j \in \mathcal{V}} w_{ij}^m ||\mathbf{u}_i - \mathbf{u}_j||_2^2.$$ 元路径实例数量

$$\mathcal{L}_2^m = \sum_{v_p \in \mathcal{V}} ||\mathbf{u}_p - \sum_{v_q \in \mathcal{N}(v_p)^m} w_{pq}^m \mathbf{u}_q||_2^2.$$

2、将静态目标转化为广义特征值求解问题：

$$(\mathbf{L} + \gamma\mathbf{H})\mathbf{U} = \mathbf{D}\Lambda\mathbf{U},$$ 二阶邻近性惩罚矩阵

3、将网络变化映射为谱扰动，增量更新嵌入：

$$\mathbf{\Lambda}^{(t+1)} = \mathbf{\Lambda}^{(t)} + \Delta\mathbf{\Lambda}, \qquad \mathbf{U}^{(t+1)} = \mathbf{U}^{(t)} + \Delta\mathbf{U}.$$

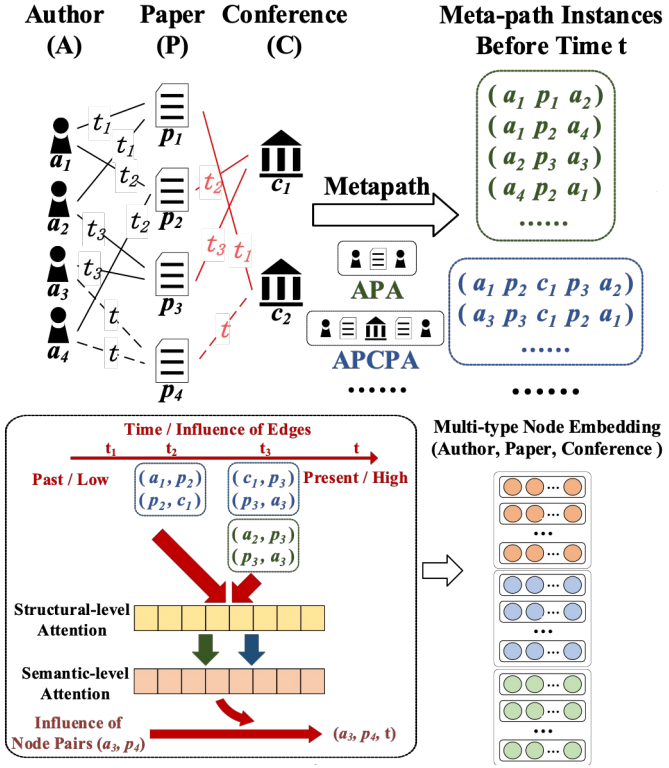
[1] Wang, Xiao, et al. Dynamic Heterogeneous Information Network Embedding With Meta-Path Based Proximity. IEEE TKDE 2022.

基于时序点过程的图嵌入

- 动机：点过程建模细粒度事件交互

$$\lambda(e) = \mu_r(u, v) + \sum_{e_h : t_h < t} \alpha(e_h, e) \cdot \kappa(t - t_h)$$

THINE[Huang2021]



1、元路径随机游走采样历史语义交互 $S(t)$

2、节点对基础影响：

$$\eta_{x,y} = -\|u_x - u_y\|^2,$$

3、历史边影响随时间减弱：

$$\kappa(t - t_{i,j}) = e^{(-\delta_i(t - t_{i,j}))}$$

节点x和y在第t时刻形成边的条件强度函数

$$\tilde{\lambda}_{x,y}(t) = \eta_{x,y} + \eta_s(t) \quad \text{元路径权重}$$
$$= \eta_{x,y} + \sum_{t_m < t} (\omega_{\mathcal{M}} \times \sum_{e_{ij} \in m} \theta_{h_{ij}} \times \eta_{e_{ij}}(t)).$$

跳数衰减

4、最大化观察边似然学习嵌入：

$$p(v_x, v_y | S(t)) = \frac{\lambda_{x,y}(t)}{\sum_{y'} \lambda_{x,y'}(t)} \quad logL = \sum_{(v_x, v_y) \in \mathcal{E}} logp(v_x, v_y | S(t))$$

[1] Hong Huang, et al. Temporal Heterogeneous Information Network Embedding. IJCAI 2021.

基于图神经网络的表征学习

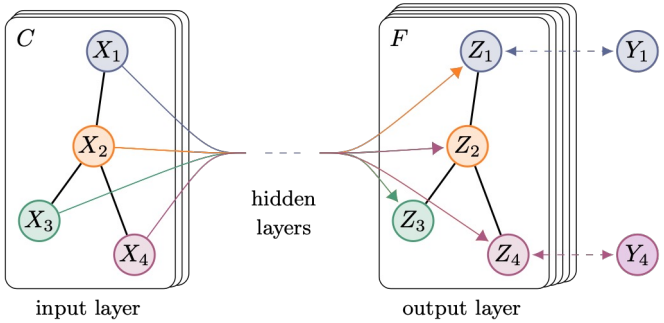
- 局部充分性：节点表征所需信息能由有限半径局部子图充分近似
- 递归可组合性：高阶结构依赖能由一跳关系递归组合表示

置换不变

消息函数 $\mathbf{m}_{i,r}^{(l)} = \mathrm{MSG}_r\left(\mathbf{h}_i^{(l)}, \left\{\mathbf{h}_j^{(l)}, \forall j \in \mathcal{N}_i^r\right\}\right),$

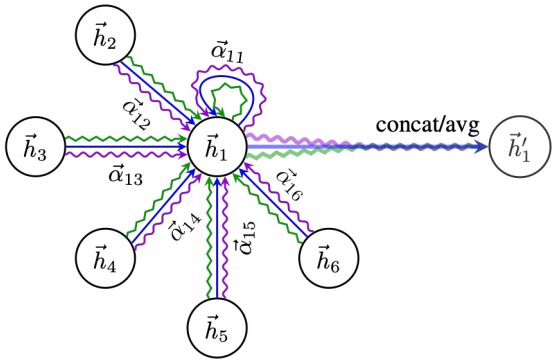
聚合更新函数 $\mathbf{h}_i^{(l+1)} = \mathrm{UPDATE}\left(\mathbf{h}_i^{(l)}, \left\{\mathbf{m}_{i,r}^{(l)}, \forall r \in \mathcal{R}_i\right\}\right),$

$$H^{(l+1)} = \sigma\left(\tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}}H^{(l)}W^{(l)}\right)$$



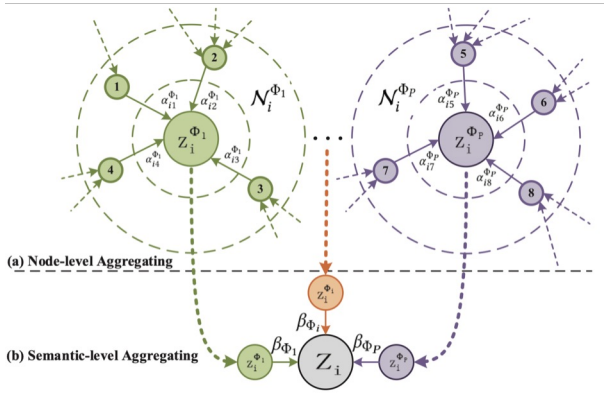
GCN[Kipf2017]

$$\alpha_{ij} = \frac{\exp\left(\mathrm{LeakyReLU}\left(\vec{\mathbf{a}}^T[\mathbf{W}\vec{h}_i \| \mathbf{W}\vec{h}_j]\right)\right)}{\sum_{k\in\mathcal{N}_i}\exp\left(\mathrm{LeakyReLU}\left(\vec{\mathbf{a}}^T[\mathbf{W}\vec{h}_i \| \mathbf{W}\vec{h}_k]\right)\right)}$$



GAT[Veličković2018]

$$w_{\Phi_p} = \frac{1}{|\mathcal{V}|}\sum_{i\in\mathcal{V}}\mathbf{q}^{\mathrm{T}}\cdot\tanh(\mathbf{W}\cdot\mathbf{z}_i^{\Phi_p}+\mathbf{b}),$$

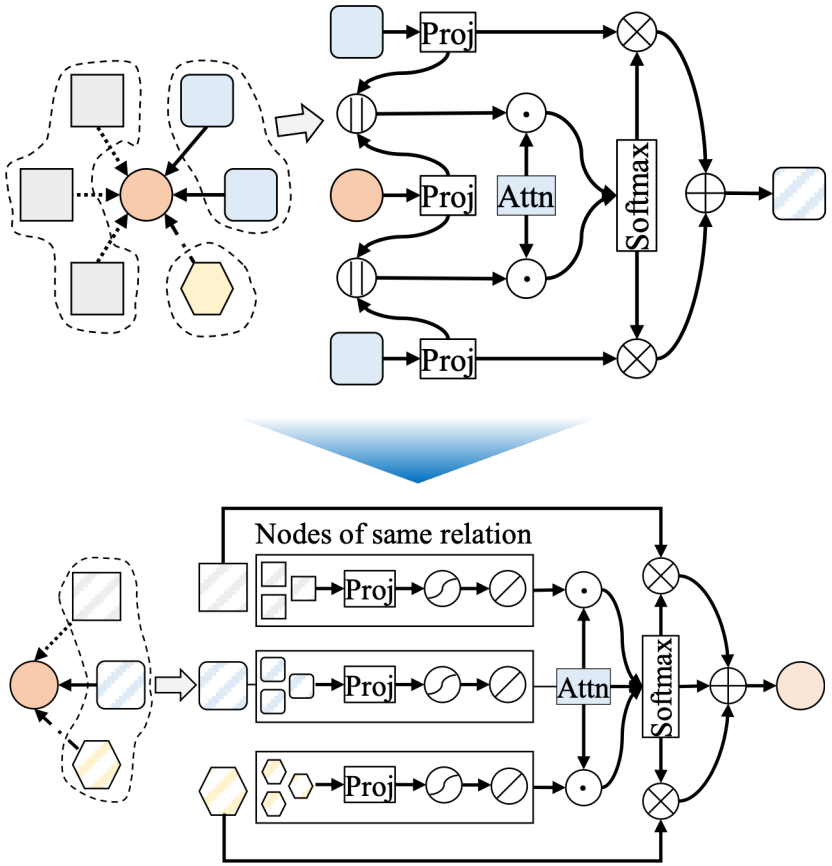


HAN[Wang2018]

基于图神经网络的表征学习

- 从空域消息传递扩展到时空异质消息传递

HTGNN[Fan2022]



DyHATR[Xue2021]

1、关系内消息传递：选择重要邻居

时间t关系r下不同邻居重要性

$$\alpha_{i,j}^{rt}=\frac{\exp(\sigma(\mathbf{a}_r^\top\cdot[\mathbf{W}^r\cdot\mathbf{x}_i||\mathbf{W}^r\cdot\mathbf{x}_j]))}{\sum_{k\in N_i^{rt}}\exp(\sigma(\mathbf{a}_r^\top\cdot[\mathbf{W}^r\cdot\mathbf{x}_i||\mathbf{W}^r\cdot\mathbf{x}_k]))}$$

关系特定表示

$$\hat{\mathbf{h}}_i^{rt}=\sigma(\sum_{j\in N_i^{rt}}\alpha_{ij}^{rt}\cdot\mathbf{W}^r\mathbf{x}_j)$$

关系特定注意力变换

2、关系间消息融合：选择重要语义

时间t下不同关系重要性

$$\beta_i^{rt}=\frac{\exp(\mathbf{q}^\top\cdot\sigma(\mathbf{W}\cdot\mathbf{h}_i^{rt}+\mathbf{b}))}{\sum_{r\in R}\exp(\mathbf{q}^\top\cdot\sigma(\mathbf{W}\cdot\mathbf{h}_i^{rt}+\mathbf{b}))}$$

快照语义表示

$$\mathbf{h}_i^t=\sum_{r=1}^{R}\beta_i^{rt}\cdot\mathbf{h}_i^{rt}$$

基于图神经网络的表征学习

3、跨时间聚合：融合多快照空域语义表征

❖ 基于循环架构的时间聚合

DyHATR[Xue2021]

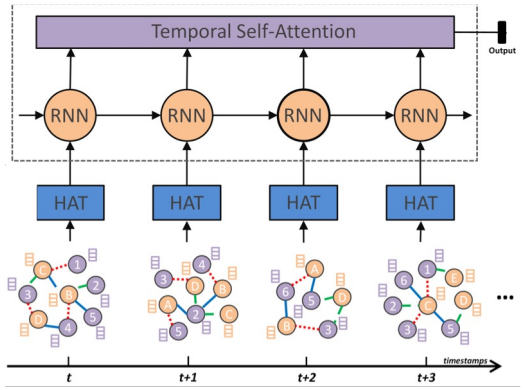
先聚合各快照表征，再用 RNN 建模节点变化

$$\mathbf{m}_v^{(t)} = f_{\text{RNN}}\left(\mathbf{m}_v^{(t-1)}, f_{\text{HGNN}}\left(G^{(t)}, \mathbf{X}^{(t)}\right)\right)$$

ReTag[Wang2024]

双GRU维护消息传递边权及表征更新

$$\mathbf{H}_{f_n(v)}^{t,l} = \sum_{r\in\mathcal{R}(v)} \overline{\alpha}_r^{t,l}(\mathbb{A}_r^{t,l}\mathbf{H}_{f_n(v)}^{t,l-1}),$$



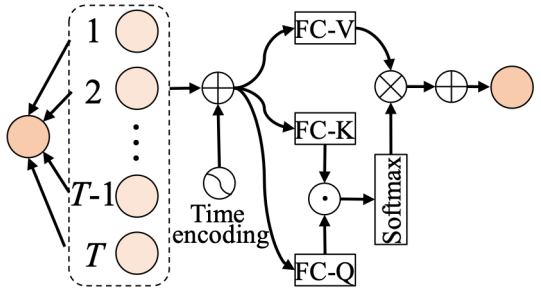
❖ 基于自注意力机制的时间聚合

HTGNN[Fan2022]

快照词元：快照序列表征

$$PE\left(\mathbf{h}_{v,R}^{t,l}\right) = \|_{i=1}^{d}\left(\mathbf{h}_{v,R}^{t,l}(i) + p(t,i)\right)$$

$$\mathbf{h}_{v,ST}^{t,l} = \sigma\left(\sum_{t'=1}^{T}\left[\gamma_v^{t',l}\right]\cdot\left[\mathbf{v}_v^{t',l}\right]\right).$$



TESA[Wang2025]

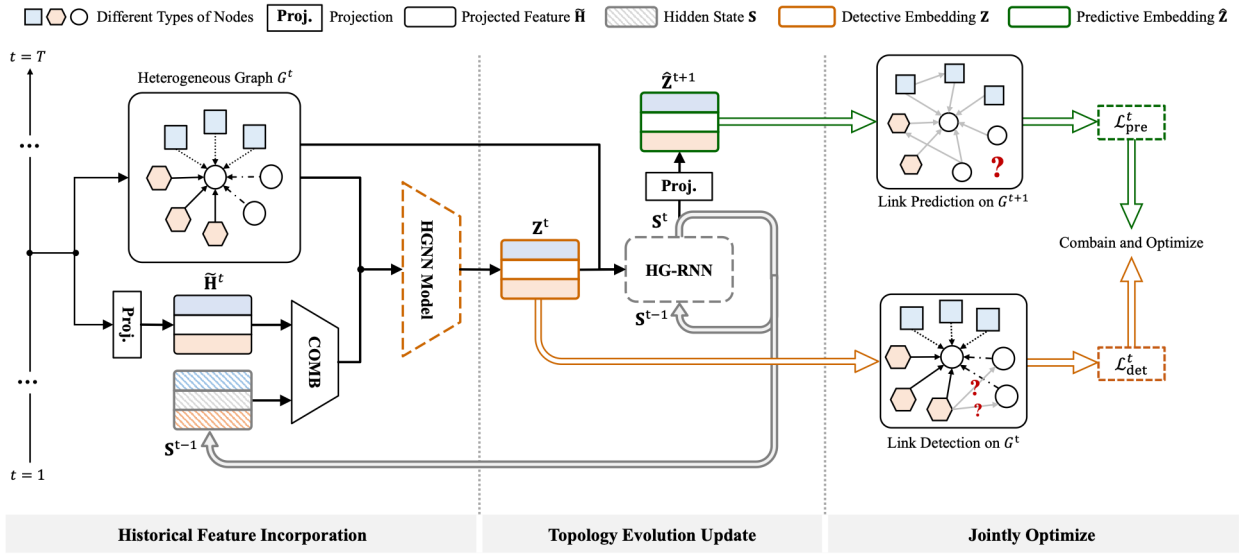
交互词元：事件轨迹表征

$$\lambda_v(t_i) = \mathbf{z}_i + \tanh\left(\sum_{e_j\in\mathcal{H}(e_i)} \frac{\mathbf{v}_j\alpha(e_j,e_i)}{1+\sum_{e_j\in\mathcal{H}(e_i)}\alpha(e_j,e_i)}\right)$$

[1] Fan Yujie, et al. Heterogeneous Temporal Graph Neural Network. SDM 2022.

基于图神经网络的表征学习

- 动态图中的演化不是节点自身的独立时间过程，而是由当前异质拓扑中多个节点和关系共同驱动的状态转移
- 如何将循环状态更新纳入消息传递过程?



HGN2T[Liu2024]

- 历史状态注入当前消息传递

$$\mathbf{h}_i^t = \text{COMB}\left(\tilde{\mathbf{h}}_i^t, \mathbf{s}_i^{t-1}\right),$$

$$\mathbf{Z}^t = \Phi\left(G^t, \mathbf{H}^t\right),$$

特定归纳偏置异质图神经网络

- 关系条件的联合状态转移

独立参数消息传递更新

$$\mathbf{U}^t = \sigma\left(\Phi_{ZU}\left(G^t, \mathbf{Z}^t\right) + \Phi_{SU}\left(G^t, \mathbf{S}^{t-1}\right)\right),$$

$$\mathbf{R}^t = \sigma\left(\Phi_{ZR}\left(G^t, \mathbf{Z}^t\right) + \Phi_{SR}\left(G^t, \mathbf{S}^{t-1}\right)\right),$$

$$\tilde{\mathbf{S}}^t = \tanh\left(\Phi_{US}\left(G^t, \mathbf{U}^t\right) + \Phi_{SS}\left(G^t, \mathbf{R}^t \odot \mathbf{S}^{t-1}\right)\right),$$

$$\mathbf{S}^t = \mathbf{U}^t \odot \mathbf{S}^{t-1} + \left(1 - \mathbf{U}^t\right) \odot \tilde{\mathbf{S}}^t,$$

基于图神经网络的表征学习

TABLE IV
AUC AND AP SCORES (MEAN ± SD %) OF THE LINK PREDICTION TASK

Model	DBLP		AMiner		Yelp		Avg. Rank
	AUC	AP	AUC	AP	AUC	AP	
VGAE	79.59 ± 3.09	84.14 ± 1.93	82.80 ± 3.43	87.75 ± 2.02	44.64 ± 3.20	47.25 ± 3.08	10.83
GATv2	81.15 ± 0.59	81.60 ± 0.82	86.11 ± 0.37	87.10 ± 0.28	55.89 ± 2.59	57.00 ± 1.46	10.17
EvolveGCN	70.50 ± 5.27	73.00 ± 4.90	75.86 ± 3.96	78.72 ± 3.96	58.58 ± 4.98	61.20 ± 5.21	12.17
VGRNN	68.65 ± 0.54	75.24 ± 0.50	73.23 ± 1.12	80.08 ± 0.83	85.72 ± 1.77	**89.78 ± 1.74**	9.17
DySAT	78.92 ± 1.32	81.06 ± 1.46	85.84 ± 1.27	87.62 ± 1.06	64.24 ± 2.02	60.91 ± 2.46	9.33
HGWaveNet	69.52 ± 2.96	73.46 ± 2.90	76.79 ± 3.33	80.31 ± 3.01	55.56 ± 1.53	56.03 ± 1.35	12.67
Mp2vec	74.23 ± 1.40	72.95 ± 1.95	80.12 ± 1.62	79.43 ± 1.95	70.05 ± 0.91	67.40 ± 0.89	10.17
HGT	69.24 ± 1.89	72.90 ± 1.89	67.64 ± 6.65	71.95 ± 5.93	61.95 ± 3.27	59.57 ± 3.28	13.67
RGCN	82.28 ± 3.79	84.18 ± 3.95	88.16 ± 1.55	89.66 ± 1.47	68.63 ± 4.53	67.30 ± 7.33	7.00
HA	86.77 ± 0.46	88.37 ± 0.32	89.88 ± 0.69	91.49 ± 0.58	62.21 ± 2.33	60.58 ± 2.20	7.00
HTGNN	84.73 ± 2.07	84.24 ± 2.40	90.61 ± 1.66	91.13 ± 1.79	63.91 ± 3.29	62.85 ± 2.11	6.67
$HGN2T_{RGCN\text{-}LSTM}$	92.80 ± 0.36	92.51 ± 0.30	96.58 ± 0.51	96.55 ± 0.46	73.70 ± 1.91	75.60 ± 1.93	4.33
$HGN2T_{RGCN\text{-}GRU}$	94.19 ± 0.19	94.17 ± 0.20	97.62 ± 0.11	97.65 ± 0.09	79.97 ± 3.46	79.65 ± 4.80	3.33
$HGN2T_{HA\text{-}LSTM}$	95.52 ± 0.25	94.86 ± 0.28	98.78 ± 0.07	98.50 ± 0.10	81.08 ± 0.80	80.88 ± 0.76	2.33
$HGN2T_{HA\text{-}GRU}$	**96.30 ± 0.22**	**95.91 ± 0.30**	**98.98 ± 0.12**	**98.79 ± 0.16**	**87.01 ± 0.83**	87.69 ± 0.89	1.17

TABLE V
AUC AND AP SCORES (MEAN ± SD %) OF THE NEW LINK PREDICTION TASK

Model	DBLP		AMiner		Yelp		Avg. Rank
	AUC	AP	AUC	AP	AUC	AP	
VGAE	54.22 ± 3.19	60.76 ± 1.88	61.83 ± 5.24	70.20 ± 3.57	43.81 ± 6.48	47.12 ± 4.61	11.67
GATv2	63.77 ± 0.59	61.39 ± 0.56	69.96 ± 0.44	69.98 ± 0.79	51.02 ± 2.94	52.23 ± 1.84	8.67
EvolveGCN	55.43 ± 1.83	55.86 ± 2.24	62.85 ± 3.00	64.22 ± 2.73	57.80 ± 5.16	62.15 ± 5.31	9.83
VGRNN	48.35 ± 0.69	52.38 ± 0.74	47.90 ± 1.63	54.52 ± 1.02	39.32 ± 0.49	52.06 ± 1.18	14.17
DySAT	57.87 ± 0.63	59.21 ± 0.93	63.52 ± 2.41	66.92 ± 1.58	49.92 ± 1.14	51.62 ± 1.59	11.17
HGWaveNet	50.90 ± 3.42	51.01 ± 2.59	58.48 ± 4.39	59.12 ± 3.34	37.90 ± 1.09	43.15 ± 0.61	14.50
Mp2vec	58.29 ± 0.50	57.70 ± 0.61	64.90 ± 2.05	64.80 ± 1.52	53.86 ± 0.62	55.81 ± 0.91	9.17
HGT	60.96 ± 2.69	62.18 ± 2.75	61.54 ± 2.80	64.41 ± 1.97	54.20 ± 0.75	53.88 ± 1.69	9.17
RGCN	59.73 ± 6.98	61.08 ± 7.46	69.03 ± 3.38	72.15 ± 3.48	51.16 ± 4.59	55.20 ± 5.91	8.33
HA	70.02 ± 0.45	69.03 ± 0.51	74.28 ± 0.84	75.64 ± 0.85	51.27 ± 1.44	52.14 ± 1.52	7.00
HTGNN	70.74 ± 2.22	68.31 ± 2.22	76.24 ± 1.93	74.71 ± 1.35	53.18 ± 3.99	54.23 ± 3.11	6.33
$HGN2T_{RGCN\text{-}LSTM}$	83.83 ± 0.68	80.65 ± 0.57	91.26 ± 0.51	90.28 ± 0.56	65.15 ± 2.58	67.48 ± 2.66	4.00
$HGN2T_{RGCN\text{-}GRU}$	86.07 ± 0.43	83.41 ± 0.52	92.95 ± 0.39	92.09 ± 0.41	66.27 ± 4.27	69.13 ± 3.83	3.00
$HGN2T_{HA\text{-}LSTM}$	89.56 ± 0.51	85.45 ± 0.69	96.77 ± 0.17	95.54 ± 0.24	71.12 ± 0.83	72.27 ± 0.58	2.00
$HGN2T_{HA\text{-}GRU}$	**90.77 ± 0.41**	**87.37 ± 0.48**	**97.06 ± 0.35**	**95.98 ± 0.47**	**72.55 ± 0.76**	**75.30 ± 0.62**	1.00

静态图神经网络方法无时序记忆性，预测比检测任务平均AUC指标下降23.39%

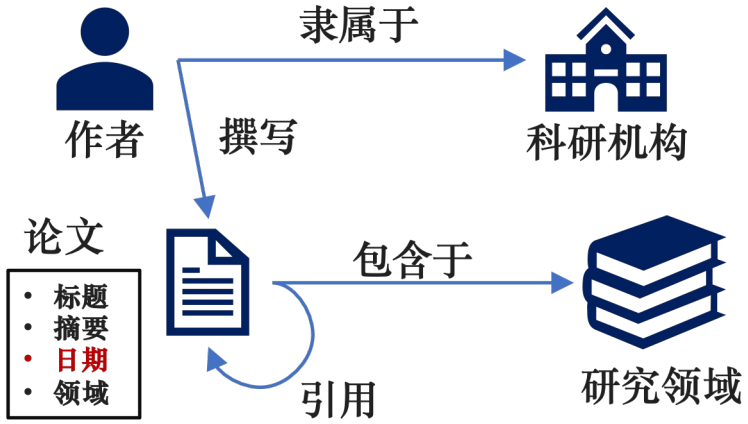
动态同质图方法由于缺失异质性建模能力，预测任务平均衰减34.66%

HGN2T联合建模降幅最低，相比最佳基线方法在三个数据集新链路预测任务上平均AUC指标 提升20.07%

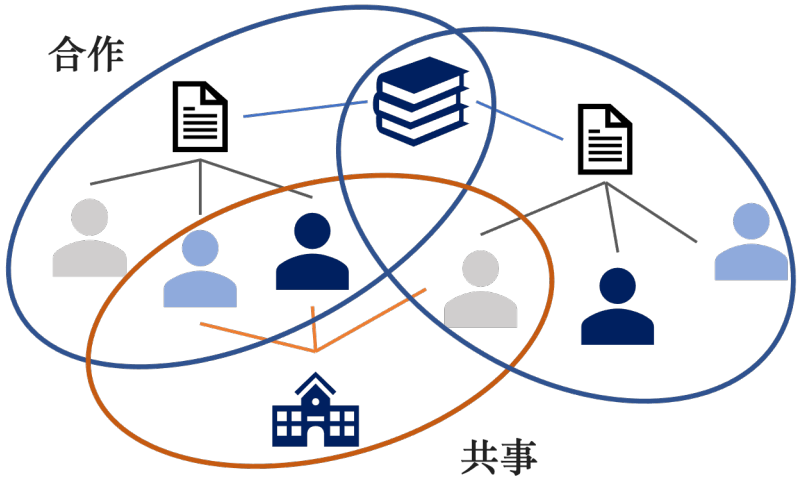
[1] Huan Liu, Pengfei Jiao*, et al. HGN2T: A Simple but Plug-and-Play Framework Extending HGNNs on Heterogeneous Temporal Graphs. IEEE Trans. on Big Data 2024.

面向高阶语义的消息传递

- 真实世界网络通常同时呈现异质性、动态性及高阶群体交互特征
- 网络底层语义蕴藏于团簇、超边等**高阶结构**之中[1]



$\mathcal{G} = (\mathcal{V}^t, \mathcal{E}^t, X^t)$
动态异质图

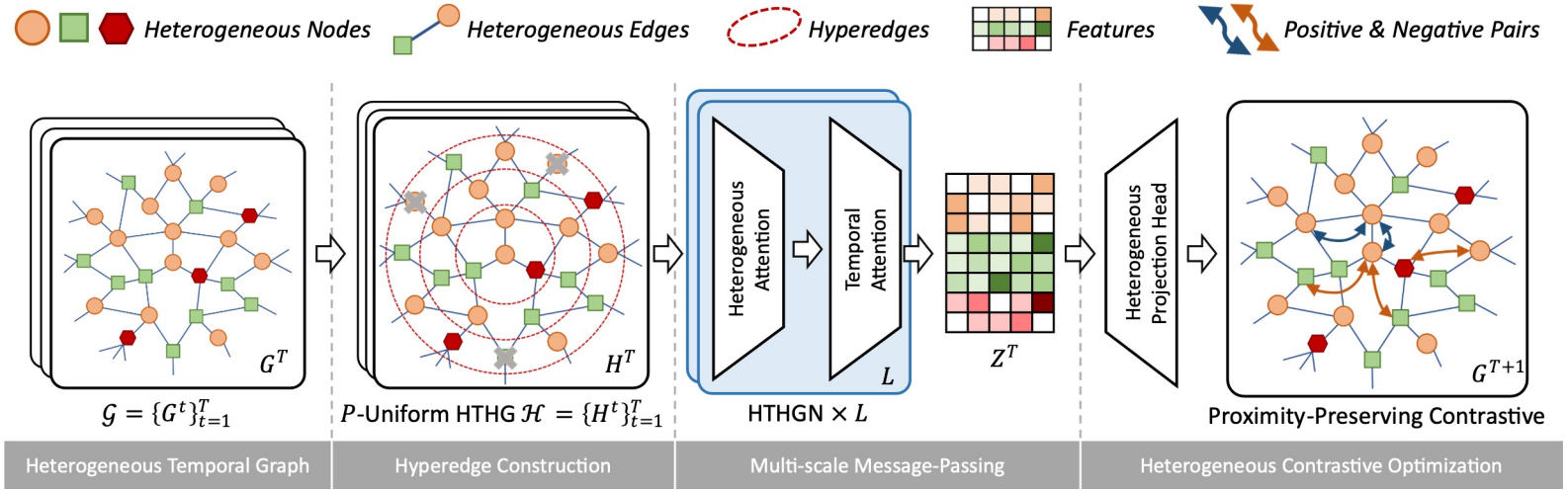


$\mathcal{H} = (\mathcal{V}^t, H^t, X^t)$
异质高阶交互感知

如何构建高阶交互感知的动态异质图通用表征预训练?

[1] Federico Battiston, et al. Networks beyond pairwise interactions: structure and dynamics. Physics Reports, 874, pp.1-92, 2020.

面向高阶语义的消息传递



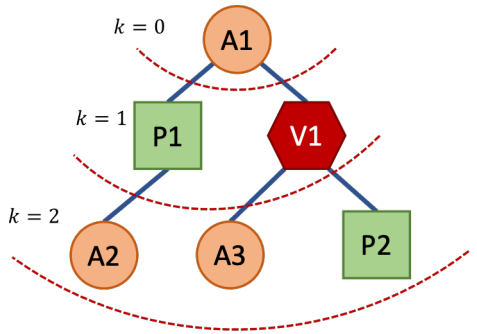
动态异质超图神经网络框架图

$$z_{i,a}^t = \Big\|_{k=1}^K \left[\sum_{j \in \mathcal{N}_i^a} \alpha_{ij}^k W_a^k z_i^t \right],$$

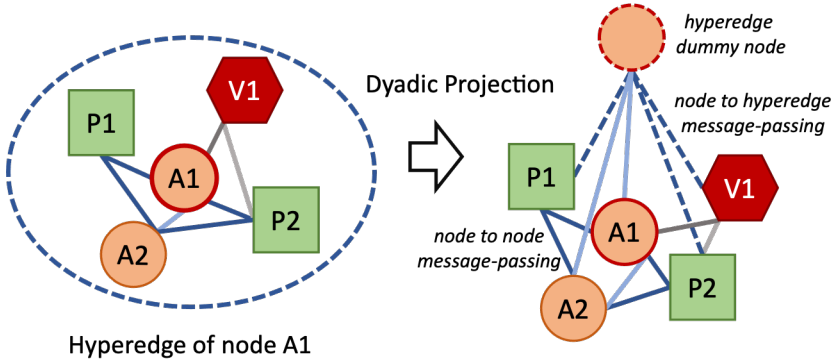
$$z_i^t = \sigma\left(\sum_{a \in \mathcal{A}_i} \beta_a z_{i,a}^t \right),$$

$$\bar{z}_i^t = \sigma\left(\mathrm{FC}\left(\sum_{t'=1}^{T} (\gamma_i^{t,t'} \cdot W_{\mathrm{V}} z_{i,p}^t) \right) \right)$$

多尺度注意力式消息传递



P-均匀异质超边构建



异质超边投影与消息传递

$$\mathbb{P}_i^{T+1} = \{u \mid u \in V^{T+1} \wedge u \in \mathcal{N}_i^{T+1}\},$$
$$\mathbb{N}_i^{T+1} = \{v \mid v \in V^{T+1} \wedge v \notin \mathcal{N}_i^{T+1} \wedge v \neq i\},$$

$$\mathcal{L} = \sum_{i \in V^t} \left(\sum_{j \in \mathbb{P}_i^{T+1}} \mathbb{E}\left[\log \mathcal{D}(\hat{z}_i, \hat{z}_j)\right] + \sum_{j \in \mathbb{N}_i^{T+1}} \mathbb{E}\left[\log\left(1 - \mathcal{D}(\hat{z}_i, \hat{z}_j)\right)\right] \right)$$

低阶结构保持的对比学习优化

[1] Huan Liu, Pengfei Jiao, et al. Heterogeneous Temporal Hypergraph Neural Network. IJCAI 2025.

面向高阶语义的消息传递

- 通过高阶语义建模能够有效提升表征学习性能

Methods	Hete.	Temp.	Hyper.
VGAE [Kipf and Welling, 2016]	✗	✗	✗
GATv2 [Brody *et al.*, 2022]	✗	✗	✗
DGI [Veličković *et al.*, 2019]	✗	✗	✗
EvolveGCN [Pareja *et al.*, 2020]	✗	✓	✗
DySAT [Sankar *et al.*, 2020]	✗	✓	✗
metapath2vec [Dong *et al.*, 2017]	✓	✗	✗
HyperGCN [Yadati *et al.*, 2019]	✗	✗	✓
UniGNN [Huang and Yang, 2021]	✗	✗	✓
HGNNP [Gao *et al.*, 2023a]	✗	✗	✓
R-GCN [Schlichtkrull *et al.*, 2018]	✓	✗	✗
HGT [Hu *et al.*, 2020]	✓	✗	✗
HetSANN-GVAE [Dalvi *et al.*, 2022]	✓	✗	✗
HPN [Ji *et al.*, 2023]	✓	✗	✗
DyHATR [Xue *et al.*, 2020]	✓	✓	✗
HTGNN [Fan *et al.*, 2022]	✓	✓	✗
HTHGN	✓	✓	✓

Table 3: Comparison of Baseline Methods in Supporting Heterogeneity (Hete.), Temporal Dynamics (Temp.), and Hyper-Relational Modeling (Hyper.)

Dataset	Yelp		DBLP		AMiner		Avg. Rank
Metrics	AUC	AP	AUC	AP	AUC	AP	
VGAE	58.62 ± 4.79	59.71 ± 4.48	77.40 ± 1.41	80.55 ± 1.53	84.56 ± 2.68	87.13 ± 2.53	9.67
GATv2	59.87 ± 1.31	57.20 ± 1.76	83.28 ± 0.13	84.71 ± 0.27	89.12 ± 0.30	90.60 ± 0.40	6.17
DGI	55.68 ± 3.11	57.36 ± 2.66	73.36 ± 2.93	77.65 ± 2.47	80.71 ± 5.59	84.04 ± 4.60	12.33
EvolveGCN	54.85 ± 5.51	54.79 ± 4.07	71.26 ± 6.87	75.33 ± 5.70	74.90 ± 7.87	78.97 ± 6.28	14.67
DySAT	61.88 ± 2.68	58.57 ± 2.71	78.61 ± 1.54	80.56 ± 1.42	83.76 ± 0.98	85.31 ± 1.18	9.50
HyperGCN	59.18 ± 2.05	55.64 ± 2.03	72.60 ± 1.04	73.96 ± 0.68	75.07 ± 1.24	75.64 ± 2.31	13.67
UniGCN	57.47 ± 5.37	54.99 ± 4.13	75.14 ± 0.76	74.45 ± 3.26	82.35 ± 0.62	81.50 ± 1.45	12.67
UniGAT	55.47 ± 0.57	52.01 ± 0.69	83.88 ± 0.31	86.54 ± 0.34	89.13 ± 0.51	91.19 ± 0.62	7.00
HGNNP	62.16 ± 3.80	60.14 ± 3.54	80.39 ± 0.20	83.39 ± 0.38	85.59 ± 0.82	88.19 ± 0.71	7.17
metapath2vec	63.79 ± 0.41	59.28 ± 0.46	64.28 ± 2.15	60.28 ± 2.56	71.05 ± 1.61	68.95 ± 1.87	13.00
R-GCN	52.72 ± 2.25	51.66 ± 1.48	82.28 ± 3.79	84.18 ± 3.95	88.16 ± 1.55	89.66 ± 1.47	9.83
HGT	55.86 ± 1.97	54.07 ± 2.00	82.32 ± 0.46	85.32 ± 0.55	87.27 ± 0.63	89.94 ± 0.53	8.33
HetSANN-GVAE	59.28 ± 2.41	57.43 ± 2.34	81.51 ± 1.53	85.08 ± 2.44	87.52 ± 0.55	90.19 ± 0.80	6.83
HPN	62.02 ± 1.04	60.24 ± 1.53	81.30 ± 1.12	82.64 ± 1.37	84.39 ± 1.06	87.97 ± 0.77	7.67
DyHATR	63.58 ± 1.37	63.60 ± 1.29	69.61 ± 1.64	69.82 ± 1.72	75.90 ± 2.51	76.80 ± 2.43	11.33
HTGNN	70.43 ± 3.36	67.45 ± 4.18	85.94 ± 3.47	87.17 ± 3.30	90.50 ± 3.33	90.81 ± 3.42	2.17
HTHGN	**74.04 ± 4.82**	**89.56 ± 3.01**	**91.33 ± 1.61**	**96.97 ± 0.56**	**96.58 ± 1.14**	**98.80 ± 0.40**	**1.00**

Table 1: AUC and AP scores of link prediction tasks between HTHGN and baselines in three datasets.

基线方法比较

相比动态异质图基线在三个数据集上平均AUC指标提升约6.2%

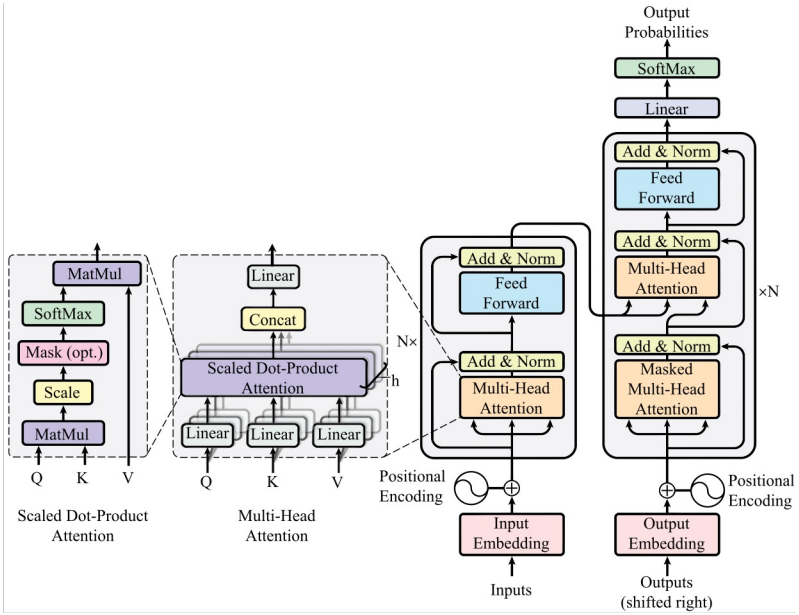
相比静态图方法AUC平均提升约9.1%

相比超图方法AUC平均提升约12.8%

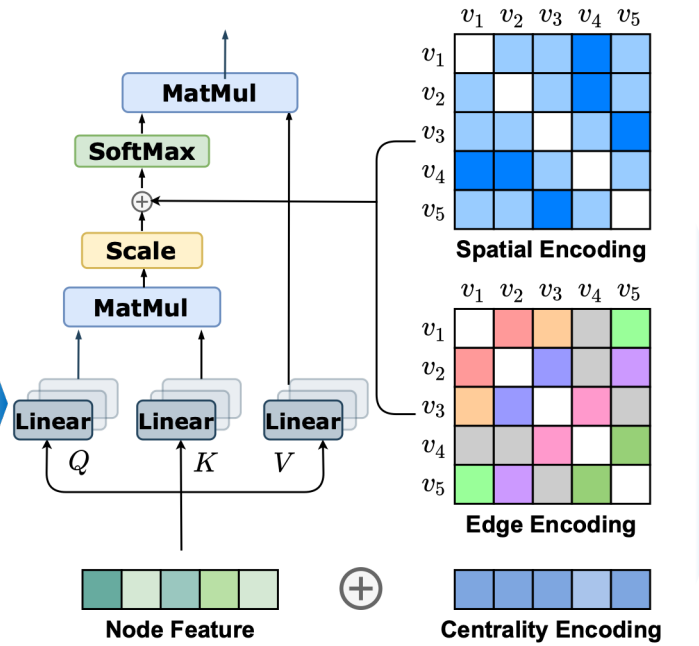
[1] Huan Liu, Pengfei Jiao, et al. Heterogeneous Temporal Hypergraph Neural Network. IJCAI 2025.

基于图Transformer的表征学习

- Transformer：目前NLP默认模型，迅速应用于视觉、语音等领域
- 核心特点：为序列而生，性能随数据量、参数量和计算量提升
- 设计空间：Tokenizing、位置编码、自注意力



Transformer[Vaswani2017]



Graphormer[Ying2021]

❖面向结构的方法
- HGT[Hu2020]
- DHGAS[Zhang2023]

❖面向交互的方法
- STHN[Li2023]
- MIGNN[Yue2025]

❖大语言模型增强的方法
- CasMLN[Wang2024]
- HTGformer[Wang2025]

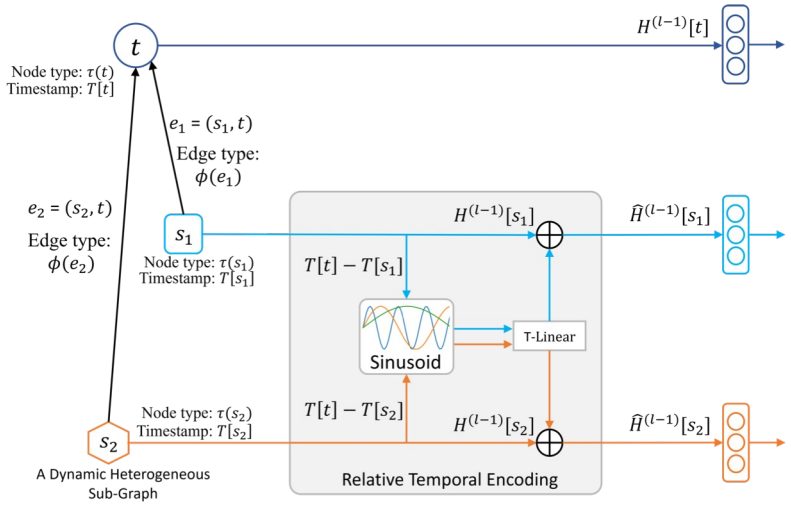
[1] Ashish Vaswani, et al. Attention Is All You Need. NeurIPS 2017.

面向结构的Transformer

- 动机：将邻域节点和边作为词元序列

$$\mathbf{h}_v^{(l)} = \sum_{u \in \mathcal{N}(v)} \alpha_{vu}^{(r)} \cdot f_{\text{MSG}}\left(\mathbf{h}_u^{(l-1)}, f_{\text{RTE}}(\Delta t)\right)$$

HGT[Hu2020]



1、元关系特定注意力权重

$$\textbf{Attention}_{HGT}(s,e,t) = \underset{\forall s \in N(t)}{\text{Softmax}}\left(\Big\|_{i\in[1,h]} ATT\text{-}head^i(s,e,t)\right)$$

$$ATT\text{-}head^i(s,e,t) = \left(K^i(s)\, W^{ATT}_{\phi(e)}\, Q^i(t)^T\right) \cdot \frac{\mu_{\langle \tau(s),\phi(e),\tau(t)\rangle}}{\sqrt{d}}$$

$$K^i(s) = \text{K-Linear}^i_{\tau(s)}\left(H^{(l-1)}[s]\right)$$

$$Q^i(t) = \text{Q-Linear}^i_{\tau(t)}\left(H^{(l-1)}[t]\right)$$

2、异质消息生成

$$\textbf{Message}_{HGT}(s,e,t) = \Big\|_{i\in[1,h]} MSG\text{-}head^i(s,e,t)$$

$$MSG\text{-}head^i(s,e,t) = \text{M-Linear}^i_{\tau(s)}\left(H^{(l-1)}[s]\right) W^{MSG}_{\phi(e)}$$

3、可学习相对位置编码

$$\widehat{H}^{(l-1)}[s] = H^{(l-1)}[s] + RTE\left(\Delta T(t,s)\right)$$

$$Base\left(\Delta T(t,s), 2i\right) = sin\left(\Delta T_{t,s} / 10000^{\frac{2i}{d}}\right)$$

$$Base\left(\Delta T(t,s), 2i+1\right) = cos\left(\Delta T_{t,s} / 10000^{\frac{2i+1}{d}}\right)$$

$$RTE\left(\Delta T(t,s)\right) = \text{T-Linear}\left(Base(\Delta T_{t,s})\right)$$

4、目标类特定注意力聚合

$$\widetilde{H}^{(l)}[t] = \bigoplus_{\forall s \in N(t)} \left(\textbf{Attention}_{HGT}(s,e,t) \cdot \textbf{Message}_{HGT}(s,e,t)\right)$$

$$H^{(l)}[t] = \text{A-Linear}_{\tau(t)}\left(\sigma\left(\widetilde{H}^{(l)}[t]\right)\right) + H^{(l-1)}[t]$$

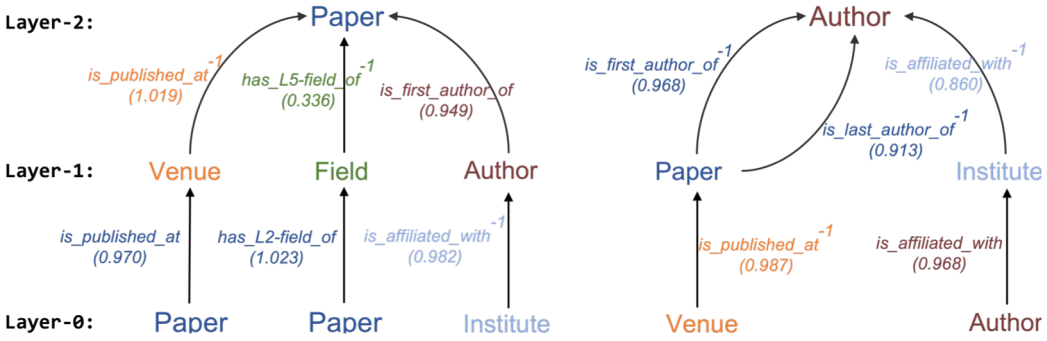
[1] Ziniu Hu, Yuxiao Dong, Kuansan Wang, Yizhou Sun. Heterogeneous Graph Transformer. WWW 2020.

面向结构的Transformer

	GNN Models		GCN [9]	RGCN [14]	GAT [22]	HetGNN [27]	HAN [23]	HGT^{-RTE}_{-Heter}	HGT^{+RTE}_{-Heter}	HGT^{-RTE}_{+Heter}	HGT^{+RTE}_{+Heter}
	# of Parameters		1.69M	8.80M	1.69M	8.41M	9.45M	3.12M	3.88M	7.44M	8.20M
	Batch Time		0.46s	1.24s	0.97s	1.35s	2.27s	1.11s	1.14s	1.48s	1.50s
CS	Paper–Field (L_1)	NDCG	.608±.062	.603±.065	.622±.071	.612±.063	.618±.058	.662±.051	.689±.042	.705±.036	**.718±.014**
		MRR	.679±.069	.683±.056	.694±.065	.689±.060	.691±.051	.751±.036	.779±.027	.799±.023	**.823±.019**
	Paper–Field (L_2)	NDCG	.344±.021	.322±.053	.357±.058	.346±.071	.352±.051	.362±.048	.371±.043	.379±.047	**.403±.041**
		MRR	.353±.053	.340±.061	.382±.057	.373±.051	.388±.065	.394±.072	.397±.064	.414±.076	**.439±.078**
	Paper–Venue	NDCG	.406±.081	.412±.076	.437±.082	.431±.074	.449±.072	.456±.069	.461±.066	.468±.074	**.473±.054**
		MRR	.215±.066	.216±.105	.239±.089	.245±.069	.254±.074	.258±.085	.265±.090	.275±.089	**.288±.088**
	Author Disambiguation	NDCG	.826±.039	.835±.042	.864±.051	.850±.056	.859±.053	.867±.048	.875±.046	.886±.048	**.894±.034**
		MRR	.661±.045	.665±.054	.694±.052	.668±.061	.688±.049	.703±.036	.712±.032	.727±.038	**.732±.038**
Med	Paper–Field (L_1)	NDCG	.560±.056	.571±.061	.584±.076	.598±.068	.607±.054	.654±.048	.667±.045	.683±.037	**.709±.029**
		MRR	.465±.055	.470±.082	.493±.069	.509±.054	.575±.057	.620±.066	.642±.062	.659±.055	**.688±.048**
	Paper–Field (L_2)	NDCG	.334±.035	.337±.051	.344±.063	.342±.048	.350±.059	.359±.053	.365±.047	.374±.050	**.384±.046**
		MRR	.337±.061	.343±.063	.370±.058	.373±.061	.379±.052	.385±.071	.397±.069	.408±.071	**.417±.074**
	Paper–Venue	NDCG	.377±.059	.383±.062	.388±.065	.412±.057	.416±.068	.421±.083	.432±.078	**.446±.083**	.445±.085
		MRR	.211±.045	.217±.058	.244±.091	.259±.072	.271±.056	.277±.081	.282±.085	.288±.074	**.291±.062**
	Author Disambiguation	MRR	.776±.042	.779±.048	.828±.044	.824±.058	.834±.056	.838±.047	.844±.041	.864±.043	**.871±.040**
		NDCG	.614±.051	.625±.049	.663±.046	.659±.061	.667±.053	.683±.055	.691±.046	.708±.041	**.718±.043**
OAG	Paper–Field (L_1)	NDCG	.508±.141	.511±.128	.534±.103	.543±.084	.544±.096	.571±.089	.578±.086	.595±.089	**.615±.084**
		MRR	.556±.136	.565±.105	.610±.096	.616±.076	.622±.092	.649±.081	.657±.078	.675±.082	**.702±.081**
	Paper–Field (L_2)	NDCG	.318±.074	.328±.046	.339±.049	.336±.062	.342±.051	.350±.045	.354±.046	.358±.052	**.367±.048**
		MRR	.322±.067	.332±.052	.348±.045	.350±.053	.358±.049	.362±.057	.369±.058	.371±.064	**.378±.071**
	Paper–Venue	NDCG	.302±.066	.313±.051	.317±.057	.309±.071	.327±.062	.334±.058	.341±.059	.353±.064	**.355±.062**
		MRR	.194±.070	.193±.047	.196±.052	.192±.059	.214±.067	.229±.061	.233±.060	.243±.048	**.247±.061**
	Author Disambiguation	NDCG	.738±.042	.755±.048	.797±.044	.803±.058	.821±.056	.835±.043	.841±.041	.847±.043	**.852±.048**
		MRR	.612±.064	.619±.057	.645±.063	.649±.052	.660±.049	.668±.059	.674±.058	.683±.066	**.688±.054**

Table 2: Experimental results of different methods over the three datasets.

- HGT相比较基线带来**9～21%**的性能提升
- 异质性和动态性不可或缺，联合建模取得最优结果
- HGT通过元关系组合能够**自适应学习**异质语义结构

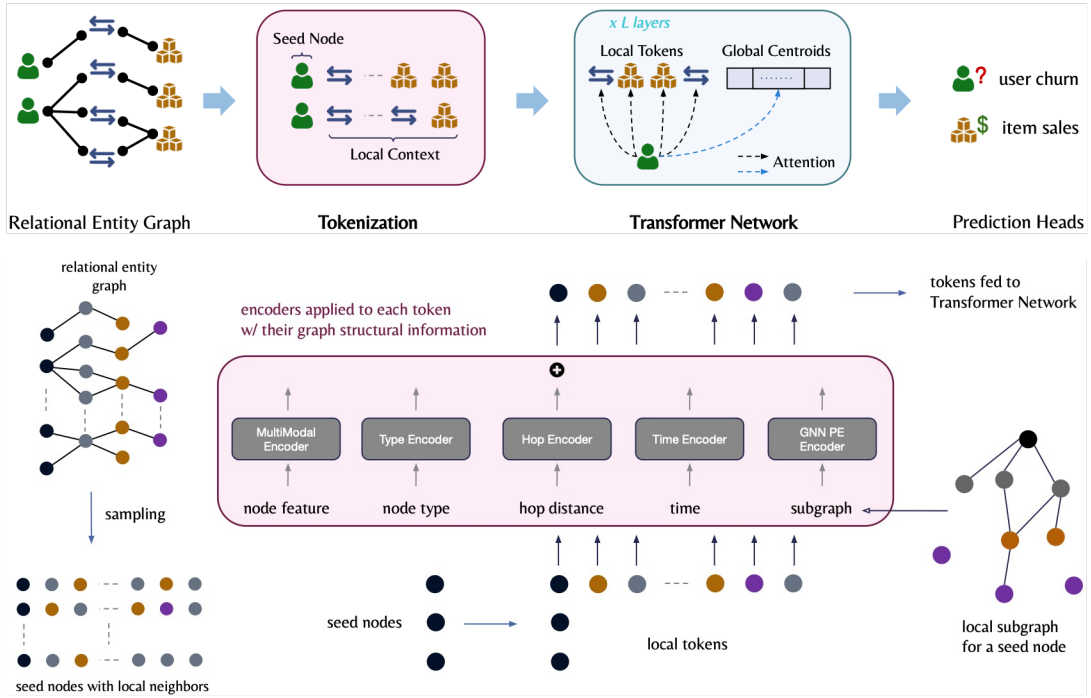


[1] Ziniu Hu, Yuxiao Dong, Kuansan Wang, Yizhou Sun. Heterogeneous Graph Transformer. WWW 2020.

面向交互的Transformer

- 动机：通过注意力机制直接建立词元间动态语义关系

RelGT[Dwivedi 2026]



0、单表记录：异质节点及属性
主外键关系：异质边及时序信息

1、局部关联：采样邻域集合，构建全对自注意力

$$h_{\text{token}}(v_j) = O \cdot [h_{\text{feat}}(v_j) \,||\, h_{\text{type}}(v_j) \,||\, h_{\text{hop}}(v_i, v_j) \,||\, h_{\text{time}}(v_i, v_j) \,||\, h_{\text{pe}}(v_j)]$$

$$h_{\text{feat}}(v_j) = \text{MultiModalEncoder}(x_{v_j}) \in \mathbb{R}^d$$

$$h_{\text{type}}(v_j) = W_{\text{type}} \cdot \text{onehot}(\phi(v_j)) \in \mathbb{R}^d$$

$$h_{\text{hop}}(v_i, v_j) = W_{\text{hop}} \cdot \text{onehot}(p(v_i, v_j)) \in \mathbb{R}^d$$

$$h_{\text{time}}(v_i, v_j) = W_{\text{time}} \cdot (\tau(v_j) - \tau(v_i)) \in \mathbb{R}^d$$

$$h_{\text{pe}}(v_j) = \text{GNN}(A_{\text{local}}, Z_{\text{random}})_j \in \mathbb{R}^d$$

2、全局关联：局部与类中心表征注意力聚合

$$h_{\text{local}}(v_i) = \text{Pool}(\text{FFN}(\text{Attention}(v_i, \{v_j\}_{j=1}^{K}))_L)$$

$$h_{\text{global}}(v_i) = \text{Attention}(v_i, \{c_b\}_{b=1}^{B})$$

$$h_{\text{output}}(v_i) = \text{FFN}([h_{\text{local}}(v_i) \,||\, h_{\text{global}}(v_i)])$$

关系数据库建模的核心在于将其重构为动态异质图

[1] Vijay Prakash Dwivedi, ..., Jure Leskovec, et al. Relational Graph Transformer. ICLR 2026.

面向交互的Transformer

❖ 异质节点分类实验（AUC↑）

Dataset	Task	RDL	HGT	HGT +PE	RelGT (ours)	% Rel. Gain
rel-f1	driver-dnf	0.7262	0.7142	0.7109	**0.7587**	4.48
	driver-top3	0.7554	0.6389	0.8340	**0.8352**	10.56
rel-avito	user-clicks	0.6590	0.6584	0.6387	**0.6830**	3.64
	user-visits	0.6620	0.6426	0.6507	**0.6678**	0.88
rel-event	user-repeat	**0.7689**	0.6717	0.6590	0.7609	-1.04
	user-ignore	0.8162	**0.8348**	0.8161	0.8157	-0.06
rel-trial	study-outcome	0.6860	0.5679	0.5691	**0.6861**	0.01
rel-amazon	user-churn	**0.7042**	0.6608	0.6589	0.7039	-0.04
	item-churn	**0.8281**	0.7824	0.7840	0.8255	-0.31
rel-stack	user-engagement	0.9021	0.8898	0.8852	**0.9053**	0.35
	user-badge	**0.8986**	0.8652	0.8518	0.8632	-3.94
rel-hm	user-churn	**0.6988**	0.6773	0.6491	0.6927	-0.87

RelGT在多数任务上优于标准局部聚合基线

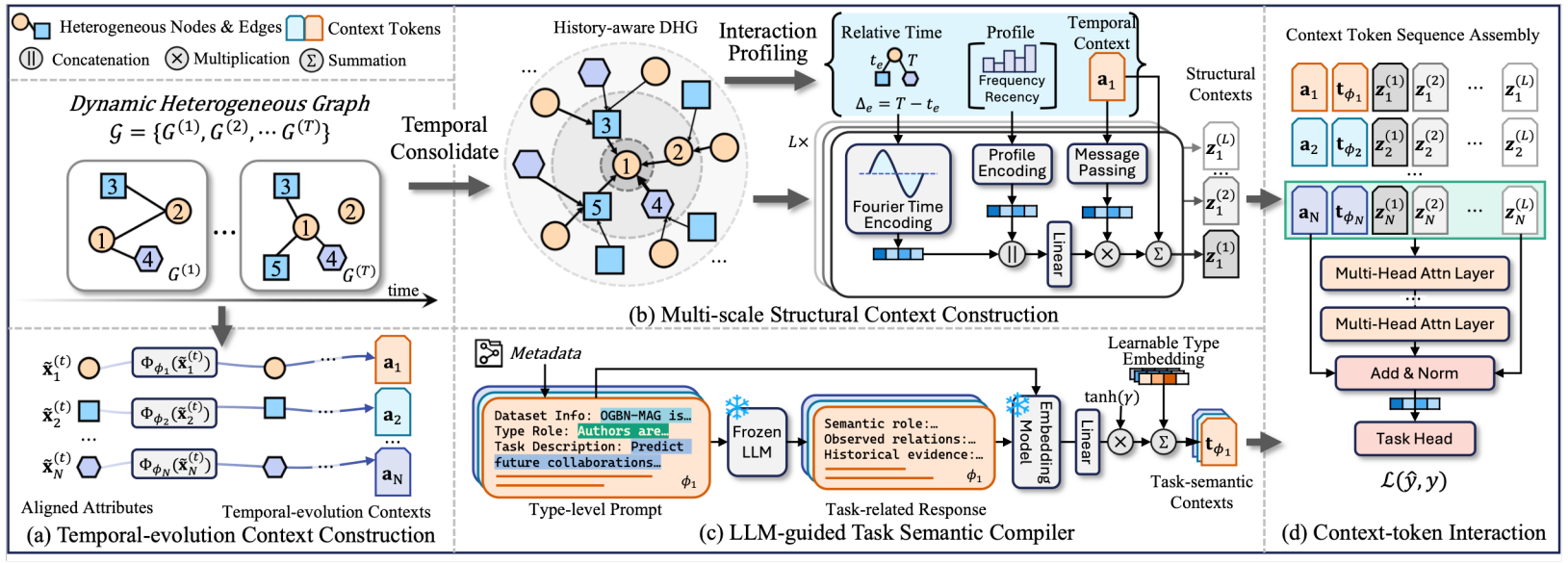
❖ 消融实验

Dataset	Task	No Global Module	No GNN PE	No Node Type	No Hop Distance	No Relative Time
rel-avito	ad-ctr	−6.00	−1.14	−7.14	−3.43	−9.14
rel-avito	user-clicks	7.85	−15.15	5.01	5.77	8.37
rel-avito	user-visits	−0.35	−2.38	−0.11	0.39	−0.75
rel-event	user-ignore	−1.30	0.12	−0.11	0.66	−0.09
rel-trial	study-outcome	−2.14	−1.72	3.74	−0.43	2.48
rel-trial	site-success	−19.01	−9.17	−2.88	−21.49	−0.71
rel-amazon	user-churn	−0.64	−0.78	0.16	0.06	−2.20
rel-hm	item-sales	−9.33	−17.35	−12.69	0.93	−77.24
	Average	−3.87	−5.95	−1.75	−2.19	−9.91

局部子图结构编码不可或缺

动态异质图Tokenization

- 主要挑战：
  - **跨时依赖保留**：如何整合分散于不同快照的异质关系，同时保留交互时序与重复模式
  - **紧凑上下文构建**：如何以规模有界的标记序列，充分表征节点演化与多尺度结构
  - **任务相关证据选择**：如何依据任务需求，自适应融合时间、结构与类型语义信息



- 编码属性轨迹，构造时变属性演化Token：

$$\mathbf{a}_v = \Phi_{\phi(v)}\left(\tilde{\mathbf{x}}_v^{(1)}, \ldots, \tilde{\mathbf{x}}_v^{(T)}\right),$$

- 整合历史拓扑，构造多尺度局部结构Token：

$$\psi(\Delta_e) = f_\theta\big([\sin(\Delta_e\boldsymbol{\omega} + \mathbf{b}) \,\|\, \cos(\Delta_e\boldsymbol{\omega} + \mathbf{b})]\big),$$

$$\bar{\mathbf{m}}_v = \frac{1}{|\mathcal{N}(v)|}\big(\mathbf{W}_{\phi(v)}\mathbf{z}_v^{(\ell-1)} + \textstyle\sum_{e\in\mathcal{N}(v)} \mathbf{m}_e\big),$$

- 数据和类型语义增广，构造任务特定语义Token：

$$\mathbf{t}_{\phi(v)} = \mathbf{e}_{\phi(v)} + \tanh(\gamma)\cdot f_p\big(\boldsymbol{\pi}_{\phi(v)}\big),$$

$$\mathbf{S}_v = \big[\mathbf{a}_v;\ \mathbf{t}_{\phi(v)};\ \mathbf{z}_v^{(1)};\ \cdots;\ \mathbf{z}_v^{(L)}\big] \in \mathbb{R}^{(L+2)\times d},$$

[1] Huan Liu, Pengfei Jiao, et al. DHGToken: Task-aware Efficient Tokenization for Dynamic Heterogeneous Graph Represent Learning.

动态异质图Tokenization

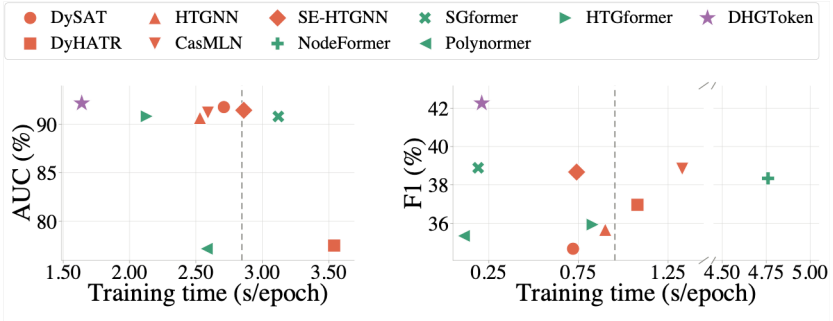
Task	Relation Prediction						Node Classification	
Dataset	MAG		Aminer		Ecomm		Yelp	
Metric	AUC	AP	AUC	AP	AUC	AP	F1	Recall
GNN-based Static Models								
GCN	86.49 ± 0.52	83.40 ± 0.59	73.67 ± 1.59	75.66 ± 1.18	76.02 ± 0.56	76.62 ± 0.58	33.77 ± 4.81	39.37 ± 2.77
GAT	88.17 ± 0.80	86.62 ± 1.49	79.48 ± 1.98	79.57 ± 1.53	74.07 ± 1.38	73.49 ± 2.02	30.59 ± 4.98	36.12 ± 3.20
RGCN	91.10 ± 1.30	88.86 ± 1.82	82.11 ± 1.34	79.45 ± 0.59	82.61 ± 0.79	82.57 ± 0.80	38.75 ± 1.06	41.71 ± 0.81
HAN	85.97 ± 3.94	84.39 ± 3.53	77.49 ± 2.90	74.08 ± 3.08	72.23 ± 4.73	70.55 ± 4.54	37.21 ± 0.39	41.51 ± 0.27
HGT	89.69 ± 0.25	87.72 ± 0.53	75.35 ± 4.77	73.16 ± 4.09	76.66 ± 3.66	73.97 ± 3.91	39.74 ± 1.24	41.74 ± 0.67
GNN-based Dynamic Models								
DySAT	91.76 ± 0.69	89.91 ± 0.84	70.83 ± 2.38	72.15 ± 2.19	72.71 ± 1.33	72.02 ± 1.40	34.67 ± 5.43	40.23 ± 3.50
HGT+	89.98 ± 0.61	88.04 ± 0.69	79.23 ± 4.08	77.29 ± 3.22	75.70 ± 1.71	76.03 ± 1.95	36.98 ± 2.56	40.64 ± 1.57
DyHATR	77.49 ± 4.67	76.86 ± 3.88	64.11 ± 4.69	66.37 ± 3.63	73.99 ± 5.44	71.85 ± 4.77	36.96 ± 0.76	41.52 ± 0.25
HTGNN	90.64 ± 0.53	88.45 ± 1.00	59.22 ± 4.20	58.33 ± 4.92	71.04 ± 6.66	69.50 ± 6.41	35.66 ± 0.69	40.50 ± 0.44
CasMLN	91.19 ± 0.85	89.95 ± 0.42	81.54 ± 3.81	79.89 ± 4.76	80.28 ± 2.05	82.15 ± 2.26	38.84 ± 1.52	41.32 ± 0.76
SE-HTGNN	91.43 ± 0.70	90.71 ± 0.56	78.97 ± 2.96	79.84 ± 2.18	81.13 ± 20.01	81.33 ± 17.30	38.67 ± 1.00	42.45 ± 0.45
Transformer-based Graph Models								
NodeFormer	OOM	OOM	75.51 ± 1.43	75.11 ± 0.97	62.45 ± 2.70	61.60 ± 2.78	38.34 ± 1.06	41.95 ± 0.70
SGformer	90.79 ± 0.45	90.09 ± 1.03	79.75 ± 1.06	80.93 ± 0.56	74.99 ± 1.06	74.05 ± 0.99	38.89 ± 0.58	41.40 ± 0.74
Polynormer	77.16 ± 1.29	74.23 ± 1.73	64.41 ± 4.84	62.58 ± 4.36	71.54 ± 2.35	69.17 ± 2.21	35.34 ± 2.72	39.44 ± 2.69
HTGformer	90.81 ± 0.79	88.48 ± 0.52	64.34 ± 6.83	64.96 ± 7.24	50.99 ± 2.29	51.58 ± 2.13	35.93 ± 0.50	40.88 ± 1.00
DHGToken	**92.16 ± 0.32**	**90.76 ± 0.70**	**83.49 ± 1.22**	**81.26 ± 1.09**	**94.85 ± 0.50**	**94.82 ± 0.80**	**42.26 ± 2.89**	**43.42 ± 1.41**
Imp. (%)	0.40	0.03	1.38	0.33	12.24	12.23	2.32	0.97

Table 1: Predictive performance on four real-world DHG datasets (*mean* ± *standard deviation*). **Bold** and underlined values denote the best and second-best results, respectively. OOM indicates that a run exceeded the available GPU memory.

在四个数据集的八项评测指标上均取得最优结果
较各指标最强基线平均提升 **3.77 个百分点**

Dataset	MAG	Aminer	Ecomm	Yelp
HGT+	9.623 (5.87)	**0.155 (0.75)**	0.545 (1.13)	0.473 (2.20)
DyHATR	3.536 (2.16)	0.750 (3.62)	0.746 (1.55)	1.076 (5.01)
HTGNN	2.533 (1.55)	0.634 (3.07)	0.741 (1.54)	0.897 (4.17)
CasMLN	2.589 (1.58)	0.806 (3.90)	1.061 (2.20)	1.331 (6.19)
SE-HTGNN	2.860 (1.75)	0.729 (3.52)	0.921 (1.91)	0.741 (3.45)
HTGformer	2.133 (1.30)	0.644 (3.11)	1.434 (2.98)	0.829 (3.86)
DHGToken	**1.639 (1.00)**	0.207 (1.00)	**0.482 (1.00)**	**0.215 (1.00)**
Imp. (×)	2.37	2.99	1.89	4.15

相较动态异质图基线单轮
训练平均加速 **2.85×**



同时获得更高预测性能与更低训练开销

[1] Huan Liu, Pengfei Jiao, et al. DHGToken: Task-aware Efficient Tokenization for Dynamic Heterogeneous Graph Represent Learning.

动态异质图表征学习：综述

- 从面向特定特征的启发式算法逐步演化至通用神经网络和Transformer架构
- 基于关系感知的图神经网络和Transformer模型得到广泛应用
- 局限：
  - 耦合建模能力仍需深化
  - 并行建模效率仍需提升
  - 通用泛化边界仍待验证

Category	Method	Granularity	Core Technique	Task	Code
Embedding-based	Change2vec [Bian *et al.*, 2019]	Dis.	Meta-path RW with Incremental Update	Clust.	Link
	THINE [Huang *et al.*, 2021]	Con.	Chronological Meta-path Walk with Hawkes Process	NC/LP/TLP	Link
	DyHINE [Xie *et al.*, 2021b]	Dis.	Relation-specific RW with Att.	TLP	-
	HPGE [Ji *et al.*, 2021]	Con.	Relation-specific Att. with TPP	NC/TLP	Link
	M-DHIN [Fang *et al.*, 2022]	Dis.	Meta-graph Sampling with LSTM	NC/LP/TLP	-
	DyHNE [Wang *et al.*, 2022b]	Dis.	Meta-path Proximity with Eigen Perturbation	NC/RP/TLP	Link
	LIME [Peng *et al.*, 2022]	Dis.	Meta-path RW with Recursive NN	NC/Clust.	Link
	SemE [Zhou *et al.*, 2023]	Con.	Semantic Meta-path Walk with Hawkes Att.	NC/RP/TLP	Link
	H2TNE [Bai *et al.*, 2022]	Con.	Relation-specific Walk with Hyperbolic Embedding	NC/LP/TLP	Link
	DHANE [Li *et al.*, 2024b]	Dis.	Meta-graph Att. with Online Update	NC	Link
GNN-based	HDGAN [Li *et al.*, 2020]	Con.	Meta-path GNN with TPP	NC/Clust.	-
	DyHAN [Yang *et al.*, 2020b]	Dis.	Relation-specific GNN with Temporal Att.	TLP	-
	DyHATR [Xue *et al.*, 2021]	Dis.	Relation-specific GNN with Temporal Att.	TLP	Link
	HTGNN [Fan *et al.*, 2022]	Dis.	Relation-specific GNN with Temporal Modeling	NR/TLP	Link
	DURENDAL [Dileo *et al.*, 2023]	Dis.	Relation-specific GNN with RNN	LP/TLP	Link
	HGN2T [Liu *et al.*, 2024]	Dis.	Relation-specific GNN with GRNN	LP/TLP	Link
	HTHGN [Liu *et al.*, 2025a]	Dis.	Relation-specific GNN with Temporal Att.	LP/TLP	Link
	TeSa [Wang *et al.*, 2025b]	Con.	Relation-specific GNN with TPP	LP/TLP	-
Transformer-based	HGT [Hu *et al.*, 2020]	Con.	Relation-specific Transformer with RTE	TLP	Link
	HT-Trans [Ruan *et al.*, 2022]	Con.	Relation-specific Transformer Encoder	LP/TLP	-
	DHGAS [Zhang *et al.*, 2023b]	Dis.	Relation- and Time-aware Self-Att.	NC/NR/RP/LP/TLP	Link
	THGAT [Zhang *et al.*, 2023a]	Con.	Relation-specific Att. with RTE	NC/Clust./TLP	Link
	STHN [Li *et al.*, 2023a]	Con.	Structural-Temporal Transformer	LP/TLP	Link
	CasMLN [Wang *et al.*, 2024a]	Dis.	LLM-enhanced Relation-specific Transformer	NC/NR/RP/LP/TLP	Link
	HTGformer [Wang, 2025]	Dis.	LLM-enhanced Transformer for DHG	NC/NR/RP/LP/TLP	-
	MIGNN [Yue *et al.*, 2025]	Con.	Att.-based GNN with GRU	LP/TLP	-
Application-oriented	HTGT [Fan *et al.*, 2021]	Dis.	Relation-specific Att. with RTE	Cybersecurity	Link
	ATHITD [Qi *et al.*, 2025]	Dis.	Relation-specific Att. with RTE	Cybersecurity	-
	HetETA [Hong *et al.*, 2020]	ST.	Heterogeneous GNN with Causal Conv.	Traffic	Link
	STHGFormer [Li *et al.*, 2024a]	ST.	Relation-specific Att. with Temporal Att.	Traffic	-
	SUPA [Wu *et al.*, 2023]	Con.	Meta-path RW with Memory Update	Recomm.	Link
	DHGP [Li *et al.*, 2024c]	Dis.	Relation- and Time-aware Att.	Recomm.	Link
	DHGRNN [Wang *et al.*, 2025a]	ST.	Relation-specific Att. with GRU	Multimodal	-
	HDGR [Dai *et al.*, 2025]	Dis.	Relation-specific GConv. with Reconstruction	Multimodal	-
	SI-HDGNN [Xu *et al.*, 2022]	Con.	Relation-specific RW with GRU	Diffusion	Link
	THGNN [Li *et al.*, 2023c]	Con.	Relation-specific Att. with LSTM	Diffusion	Link

Table 1: Comparison of DHG representation learning methods. Task abbreviations: NC (node classification), LP/TLP (link / temporal link prediction), RP (relation prediction), NR (node regression), Clust. (node clustering). Temporal granularity: Dis. (discrete-time), Con. (continuous-time), ST. (spatio-temporal). Technique abbreviations: RW (random walk), Att. (attention), GConv. (graph convolution), GRNN (graph recurrent neural network), TPP (temporal point process), RTE (relative time encoding).

[1] Huan Liu, Pengfei Jiao*, et al. Dynamic Heterogeneous Graph Representation Learning: A Survey. IJCAI 2026.

总结与展望

- 在本次报告中：
  - 梳理动态异质图表征学习发展脉络，构建多类方法的统一技术体系
  - 比较动态异质图多类方法建模机制，归纳技术路线优势与适用边界
  - 总结动态异质图数据评估基准现状，提炼方法共性规律与关键挑战
- 展望：面向真实网络统一表征的图基础模型
  - 可生成：如何实现动态异质图生成式预训练？
  - 可泛化：如何构建多场景任务通用泛化适配？
  - 可解释：如何挖掘复杂性实现可解释图推理？

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